Energy-Efficient 200 Gbit/s Transceivers for Optical Access Networks

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Chapter 1

Access Networks and Coherent PON

1.1 Optical Access Networks

The access network is the final link between an operator's central exchange and the end user, often called the *last mile*. Exponentially growing bandwidth demands from streaming, cloud services, and 5G backhaul are placing severe pressure on this segment. Traditional intensity-modulation direct-detection (IM-DD) passive optical networks, standardised through the ITU-T and IEEE, are approaching their practical limits at data rates approaching 50 Gbps per wavelength [1].

A passive optical network (PON) uses a point-to-multipoint (P2MP) topology in which a single optical line terminal (OLT) at the exchange serves multiple optical network units (ONUs) at the customer premises through a tree of passive optical splitters, as illustrated in fig. 1.1. No active components are required in the distribution plant, keeping operational costs low.

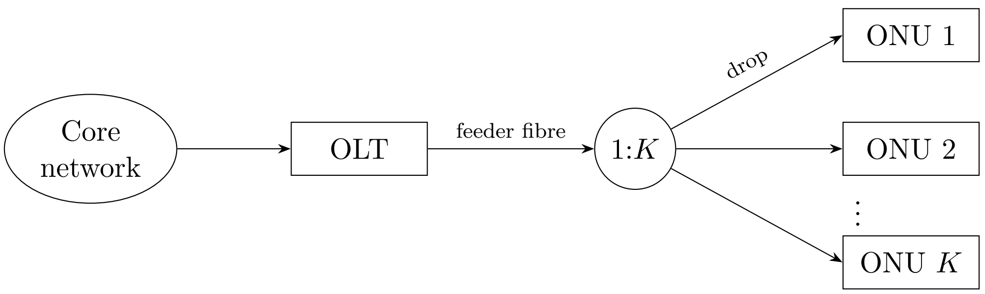


Figure 1.1: Passive optical network: one OLT serves K ONUs via a passive 1:K splitter.

1.2 Coherent PON

Coherent optical technology encodes information in the amplitude, phase, and polarisation of the optical carrier. Mixing the received signal with a local oscillator (LO) laser at the receiver provides coherent gain that dramatically improves receiver sensitivity relative to direct detection, partially recovering the power lost at the splitter. Linear transmission impairments — chromatic dispersion (CD) and polarisation-mode dispersion (PMD) — can be corrected entirely in DSP, removing the need for dispersion-compensating fibre.

These properties make coherent optics an attractive basis for the next generation of high-split, long-reach PON.

The Coherent Passive Optical Network (CPON) specification [1] defines a 100 Gbps single-wavelength P2MP system. This report focuses on the *downstream* direction (OLT to ONU); the key physical-layer parameters are described in the following sections.

1.3 Modulation and Symbol Rate

The downstream signal uses non-differential DP-QPSK at a symbol rate of $R_s = 30.5\,\text{GBd}$ ($\pm 20\,\text{ppm}$), giving a PMD-layer line rate of $4R_s \approx 122\,\text{Gbps}$. Each polarisation carries 2 bits per symbol by mapping consecutive bit pairs to in-phase and quadrature amplitudes $I, Q \in \{-1, +1\}$, giving QPSK constellation points at $\pm 1 \pm j$. Training and pilot symbols use the same mapping but at amplitude ± 3, placing them outside the data constellation for unambiguous identification.

1.4 Downstream Frame Structure

The fundamental framing unit is the *DSP subframe* of 3 712 symbols, organised as 116×32-symbol blocks. Every block begins with a pilot symbol; the first block also carries an 11-symbol training sequence, whose first symbol doubles as pilot index 1.

1.5 Key System Tolerances

The OLT transmitter laser must be within $\pm 1.5\,\text{GHz}$ of its DWDM channel centre, and the ONU must acquire signals within $\pm 1.8\,\text{GHz}$ of the grid. In the worst case, the combined transmitter and LO offsets yield a carrier frequency offset (CFO) of up to $\approx 3\,\text{GHz}$ that the receiver DSP must correct. Both OLT and ONU lasers must have a linewidth of at most 1 MHz; the combined linewidth drives the phase noise that the carrier recovery block must track.

The specification requires a pre-FEC bit error ratio of $\leq 2 \times 10^{-2}$, which the hard-decision LDPC FEC reduces to a post-FEC BER of $\leq 10^{-12}$.

Chapter 2

Experimental Setup and Methodology

2.1 Simulation Framework

All DSP algorithms in this report are designed, characterised, and compared through software simulation in MATLAB. The simulated receiver processes DP-QPSK symbols generated at the CPON downstream parameters described in chapter 1: symbol rate $R_s = 30.5\,\mathrm{GBd}$, with pilot and training symbols embedded in the subframe structure of section 1.4. The channel model adds impairments such as chromatic dispersion, polarisation-mode dispersion, AWGN, phase noise, carrier frequency offset to test the performance of the algorithms.

2.2 Fixed-Point Arithmetic

Real hardware DSP implementations operate on fixed-point integers rather than floating-point numbers. Representing a value with n bits limits both the dynamic range and resolution of arithmetic, and the energy cost of addition and multiplication scales with n (see section 2.4). Choosing the minimum n that still meets the pre-FEC BER constraint is therefore central to minimising receiver energy.

Fixed-point behaviour is modelled using MATLAB's `fi` (fixed-point) object, which performs bit-accurate integer arithmetic with configurable overflow and rounding modes.

2.3 Energy Estimation

The energy efficiency of a DSP technique used in a coherent receiver cannot be judged from the receiver alone: a simpler implementation that saves receiver energy typically requires a higher SNR, which the transmitter must compensate for with extra launch power. A like-for-like comparison between architectures must therefore account for both contributions.

The receiver contribution is modelled from operation counts of real additions and multiplications, with per-operation energies that depend on the fixed-point word length, allowing implementations to be compared without a hardware build. The transmitter contribution is derived from the minimum SNR each implementation requires to meet the pre-FEC BER threshold defined in the CPON standard [1]. The two are combined for a downstream PON to give the per-bit system energy.

2.4 Receiver Energy

The energy consumption of the receiver can be estimated from the number of real multiplications N_M and real additions N_A it performs to process each symbol. The true cost of these operations depends on the hardware implementation. Here the reference values shown in table 2.1 are used to allow for comparison between different DSP algorithms. An accurate estimate of their actual energy consumption would require analysis of their ASIC implementation for a modern process node.

Table 2.1: Energy cost of arithmetic operations for different bit widths in 45 nm CMOS [4].

n (bits)	Add (pJ)	Multiply (pJ)
8	0.03	0.2
32	0.1	3.1

2.4.1 Dependence on Word Length

A straightforward adder implementation performs n bit-wise additions on words of length n, so its energy scales linearly with n. Multiplication accumulates n partial products of length n and so scales as n^2. These physical scaling laws are imposed *a priori* and the proportionality constants calibrated by least-squares fit through the origin to the two data points of table 2.1, giving

$$E_A(n) = 3.16\,n \text{ fJ}, \qquad E_M(n) = 3.03\,n^2 \text{ fJ}. \tag{2.1}$$

2.4.2 Amortisation

Some DSP algorithms perform computations that are reused over multiple symbols. For example, carrier frequency offset (CFO) varies much slower than the symbol rate so can be estimated less frequently. For a computation that requires N_M multiplications and N_A additions and is used by S symbols, the operations per symbol would be N_M/S and N_A/S.

2.4.3 Total Energy and Power

The energy per symbol for the entire receiver is given by

$$E_{\mathrm{rx},s} = \sum_{k=1}^{L} 2\left[N_{A,k}E_A(n_k) + N_{M,k}E_M(n_k)\right], \tag{2.2}$$

where the receiver is decomposed into L blocks that use different word lengths n_k according to their required precision, and $N_{A,k}$, $N_{M,k}$ are the addition and multiplication counts per symbol. The factor of 2 is for both polarizations. Blocks operating on the oversampled signal at rate $f > R_s$ have their per-sample operation counts scaled by the oversampling ratio f/R_s. The receiver power is then

$$P_{\mathrm{rx}} = R_s E_{\mathrm{rx},s}, \tag{2.3}$$

and converting to energy per bit allows fair comparison across modulation schemes and symbol rates,

$$E_{\mathrm{rx}} = \frac{E_{\mathrm{rx},s}}{\log_2(M)}, \tag{2.4}$$

where M is the modulation order. Complex multiplications and additions are needed in many DSP algorithms. 4 real multiplications and 2 real additions are assumed for each complex multiplication, and 2 real additions for complex addition.

2.5 Transmitter Energy

The downstream link considered here consists of a single OLT transmitting through a 1:K passive splitter to K ONUs, as shown in fig. 2.1. The OLT power is divided equally between branches, so each ONU receives a $1/K$ share. This is a simplification of a typical PON where there may be several stages of passive splitters placed at different locations along the fibre, in which case the noise characteristics of the fibre between splitter stages must be analysed independently.

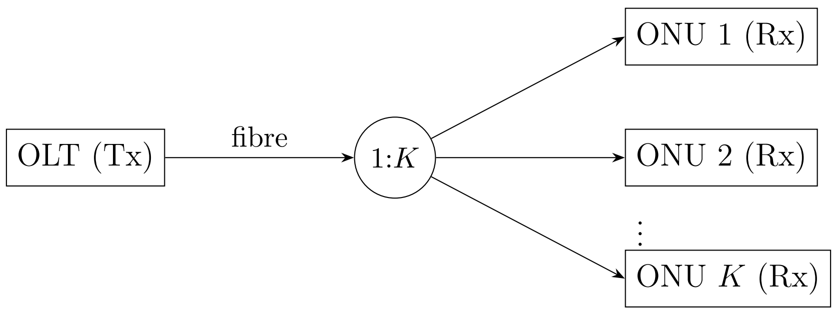


Figure 2.1: Downstream PON link: one OLT transmitting to K ONUs.

2.5.1 Minimum Required SNR

The CPON standard specifies a maximum pre-FEC bit-error rate (BER) of 10^{-2} that must not be exceeded for the forward error-correction (FEC) scheme to function correctly. Each receiver implementation therefore has a minimum required SNR to meet this FEC threshold. This SNR, together with the noise characteristics of the optical network, determines the minimum transmitter power.

2.5.2 Noise

Shot Noise

The relationship between SNR and transmitter power P_{tx} depends on the dominant source of noise in the network. Optical access networks typically operate over short lengths at low power, so amplifiers can be omitted and non-linear noise neglected, leaving shot noise as the dominant impairment. The shot noise power at the receiver is $2h\nu B_{\text{eff}}$, where B_{eff} is the bandwidth of the photodetector; this can be matched to the bandwidth of the received signal after the 1:K splitter in fig. 2.1 with a bandpass filter. The received power after attenuation and the splitter (assuming equal power per channel) is $P_{\text{tx}}\exp(-\alpha L)/K$, giving a per-ONU SNR of

$$\text{SNR} = \frac{P_{\text{tx}}\exp(-\alpha L)}{2Kh\nu B_{\text{eff}}}, \tag{2.5}$$

where α is the fibre loss in nepers/km , and L is the fibre length in km. Rearranging gives the transmitted power as

$$P_{\text{tx}} = 2Kh\nu B_{\text{eff}}\exp(\alpha L)\cdot\text{SNR}, \tag{2.6}$$

which is proportional to K when channels transmit equal power. The energy per bit transmitted is then

$$E_{\text{tx}} = \frac{P_{\text{tx}}}{KR_s\log_2(M)}. \tag{2.7}$$

Amplifier and Non-Linear Noise

Long-reach access networks require EDFAs in the trunk fibre, in which case amplifier and non-linear noise can no longer be neglected. Each EDFA with gain G and noise figure NF (dB) operating over a full WDM bandwidth B contributes an ASE noise power

$$P_{\text{ASE}} = N_{\text{ASE}}B = 10^{\text{NF}/10}h\nu(G-1)B, \tag{2.8}$$

and assuming each amplifier exactly compensates the loss of its preceding span, every span re-launches with the same input power P_{tx}. The Gaussian Noise model [6] then gives the per-span non-linear interference (NLI) power as

$$P_{\text{NLI}} = \frac{C_{\text{NLI}}P_{\text{tx}}^3}{B^2}, \tag{2.9}$$

where $C_{\text{NLI}} = \frac{8\gamma^2 L_{\text{eff}}^2\alpha}{27\pi|\beta_2|}\ln(\frac{|\beta_2|}{\alpha}\pi^2B^2)$ depends on the fibre and link parameters.

For a link of n identical spans, the ASE and NLI contributions from each span accumulate, giving a total noise power $n(P_{\text{ASE}} + P_{\text{NLI}})$. With the 1:$K$ passive splitter placed at the end of the trunk, both signal and accumulated noise are attenuated equally between the OLT and each ONU, so the splitter ratio cancels in the per-ONU SNR:

$$\text{SNR} = \frac{P_{\text{tx}}}{n\left(N_{\text{ASE}}B + \frac{C_{\text{NLI}}P_{\text{tx}}^3}{B^2}\right)}, \tag{2.10}$$

where P_{tx} is both the OLT transmit power and (under the equal-compensation assumption) the launch power into each span. The required transmit powers for each implementation are solved for numerically from eq. (2.10), with the transmitter energies following from eq. (2.7). Here the transmitted power is not proportional to K for a given SNR even if channels transmit equal power due to the dependence of C_{NLI} on the bandwidth.

2.6 System Energy

The total energy consumed per bit for network is given by

$$E_{\text{sys}} = \frac{1}{\eta} E_{\text{tx}} + E_{\text{rx}}, \tag{2.11}$$

where η is the wall-plug efficiency of the OLT laser. In the shot noise regime, E_{sys} is independent of K, but will vary when amplifier and non-linear noise dominate.

Chapter 3

Carrier Recovery

3.1 Introduction

A coherent receiver recovers the transmitted symbols by mixing the incoming optical signal with a local oscillator (LO) laser, allowing the amplitude and phase of each symbol to be extracted directly. However, the transmitter and LO lasers will in general have different centre frequencies and independent phase trajectories, producing a carrier frequency offset (CFO) and accumulated phase noise on the received symbols. Carrier recovery is the DSP stage responsible for estimating and removing these impairments before symbol decisions are made.

Carrier recovery is divided into two sequential stages. Frequency recovery performs a coarse estimate of the CFO and corrects it so that only slow residual phase variations remain. Phase recovery then tracks and removes these residual phase errors on a block-by-block basis. The algorithm chosen for each stage determines both the minimum achievable SNR and the receiver DSP energy, making algorithm selection central to the system-level energy comparison of chapter 2.

This chapter describes the Rife and Boorstyn (R&B) and Differential Kay (DK) frequency recovery algorithms and the Viterbi & Viterbi (V&V) and pilots-only phase recovery algorithms, analysing their estimation accuracy and computational complexity. The optimal fixed-point word length for each combination is then found, and the resulting system energies are compared across shot-noise and amplified link scenarios.

3.2 Frequency Recovery

3.2.1 Isolating CFO

All frequency recovery algorithms first require that the phase of the data be removed before estimation. For QPSK symbols $c[i] = e^{j\frac{\pi}{4}+m[i]\frac{\pi}{2}}$, this can be done by raising each received symbol $x[i] = c[i]e^{j(2\pi f_d i+\theta[i])} + n[i]$ to the fourth power to give

$$z[i] = x^4[i] = e^{j(8\pi f_d i+4\theta[i])} + n'[i] \tag{3.1}$$

where θ is phase noise, f_d is the CFO, and n is additive noise. This allows all data symbols to be used in CFO estimation, but amplifies phase and additive noise, which degrades the estimation quality. The fourth power operation also requires three complex multiplications. An alternative is to use pilot symbols where the data phase is known beforehand, giving

$$z[i] = x[i]c^*[i] = e^{j(2\pi f_d i+\theta[i])} + n[i], \tag{3.2}$$

which is more robust to noise and less computationally expensive. Some algorithms only require $\arg(z[i])$, which is efficiently found using CORDIC as

$$\arg(z[i]) = \{4\arg(x[i])\}|_{-\pi}^{\pi} \tag{3.3}$$

for blind operation or

$$\arg(z[i]) = \{\arg(x[i]) - \arg(c[i])\}|_{-\pi}^{\pi} \tag{3.4}$$

for data-aided operation, where the wrap operation can be done with no additional cost by aligning the wrap range to the overflow range of the fixed-point value used to store the phase.

3.2.2 Rife and Boorstyn

The value of the CFO will appear as the strongest peak in the frequency spectrum of $z[i], z[i+1], \ldots, z[i+L]$ where L is the observation length. The Rife and Boorstyn (R&B) algorithm first performs an FFT on the observed sequence and finds the index k corresponding to the strongest peak. This coarse estimation is refined with a parabolic interpolation on the complex FFT values X_{k-1}, X_k, X_{k+1} [5] to give

$$\delta = -\Re\left[\frac{X_{k+1} - X_{k-1}}{2X_k - X_{k-1} - X_{k+1}}\right]. \tag{3.5}$$

The updated index $k' = k + \delta$ is then scaled by the sampling rate to find the true CFO. The resolution in k is $\epsilon(k) = 1/2N_{FFT}$. The CPON specification provides only $L = 11$ training symbols, so data-aided operation without zero-padding gives $\epsilon(\hat{f}_d) \approx 0.05\,R_s$ — too coarse for the downstream phase recovery to correct. Zero-padding the training sequence to a larger N_{FFT} improves the resolution without requiring additional training data. The normalised mean squared-error (MSE) in the estimated CFO is shown in fig. 3.1 for data-aided operation with 11 training symbols. A CFO estimate is computed for both polarizations and averaged to reduce the MSE. The modified Cramer-Rao bound (MCRB) [3] provides a lower bound on the achievable MSE and is given by

$$\text{MCRB} = \frac{3}{2\pi^2 L^3 \times \text{SNR}}. \tag{3.6}$$

R&B achieves this bound above a threshold SNR of ≈ 4 dB, where the threshold SNR is defined as the SNR below which the estimator begins to deviate from the MCRB. Below this, the noise has a significant contribution to the FFT values and can lead to an incorrect index being chosen.

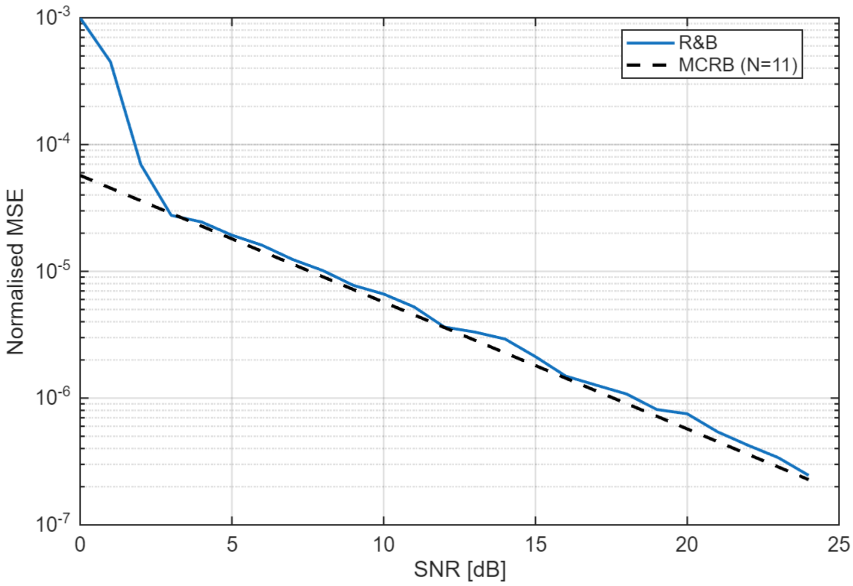


Figure 3.1: Normalised MSE of the R&B algorithm with FFT size $N_{FFT} = 512$ compared to the MCRB for $L = 11$ training symbols.

Data-aided operation is constrained by the length of the training sequence, but the accuracy of the R&B algorithm improves with longer observation lengths, as shown in fig. 3.2 for non-data-aided operation. The SNR threshold at which this algorithm begins to deviate from the MCRB is initially higher with data-aided operation due to the amplification of noise, but decreases for longer observation lengths.

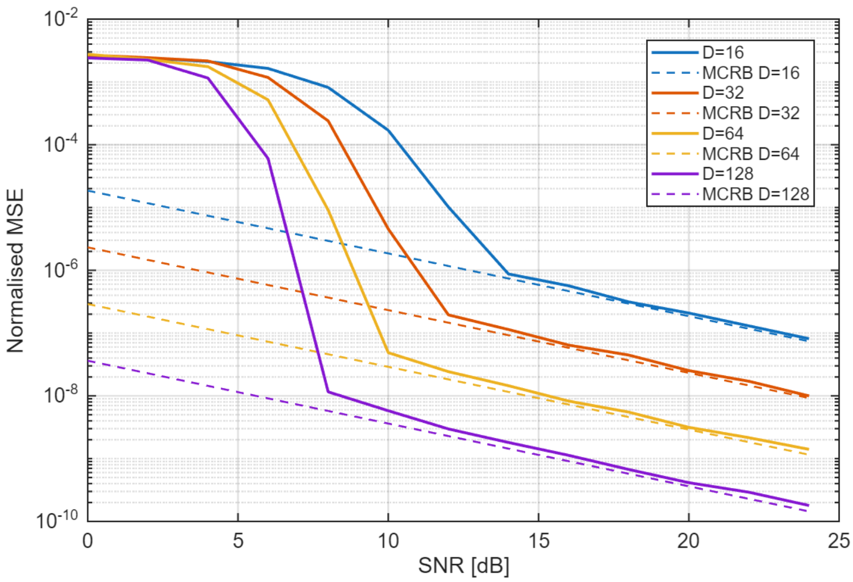


Figure 3.2: Normalised MSE of the R&B algorithm for non-data-aided operation and different observation lengths.

3.2.3 Differential Phase + Kay Estimator

The R&B algorithm exhibits good performance even at low SNRs, but is computationally expensive. The FFT requires $\frac{N_{FFT}}{2}\log_2(N_{FFT})$ complex multiplications and

$N_{FFT} \log_2(N_{FFT})$ complex additions. The fine search from eq. (3.5) also requires a complex division. A cheaper approach would involve computing the phase difference

$$\arg(z[i]) - \arg(z[i-1]) = 2\pi f_d T_s + \theta[i] - \theta[i-1] + \eta[i] \tag{3.7}$$

over several symbols and averaging to remove the effect of θ. Estimating f_d in this manner is vulnerable to the effect of additive noise on the phase $\eta[i]$, so exhibits poor accuracy. This can be improved with a second pass using the Kay estimator. The Kay estimator uses a least-squares approach, computing the estimated CFO as

$$\hat{f}_d = \frac{R_s}{2\pi} \sum_{k=1}^{L} \frac{6k(L-k)}{L(L^2-1)} \arg(z[k]z^*[k-1]). \tag{3.8}$$

where $\arg(z[k]z^*[k-1])$ computes the phase difference from eq. (3.7) and $\frac{6k(L-k)}{L(L^2-1)}$ is a weighting term that prioritises the contribution of phase differences in the middle of the observed sequence. This estimator is able to achieve the MCRB, but the least-squares approach considers phase differences of non-adjacent symbols, making it vulnerable to phase wrapping that distorts the estimate when the CFO is high enough. When the total phase drift $\Delta\phi = 2\pi f_d L T_s$ across the observation window exceeds 2π, a wrap shifts the estimated CFO by $\pm 1/(LT_s)$. A first-pass coarse CFO correction must therefore be applied before the Kay estimator to keep the residual CFO within the unambiguous range. The performance of the Differential Kay (DK) estimator for data-aided operation is shown in fig. 3.3. The MCRB is achieved above a threshold SNR of ≈ 7 dB. This is higher than for R&B, but at a reduced computational cost. Unlike the R&B algorithm, the threshold SNR of the DK estimator does not depend on observation length, as shown in fig. 3.4.

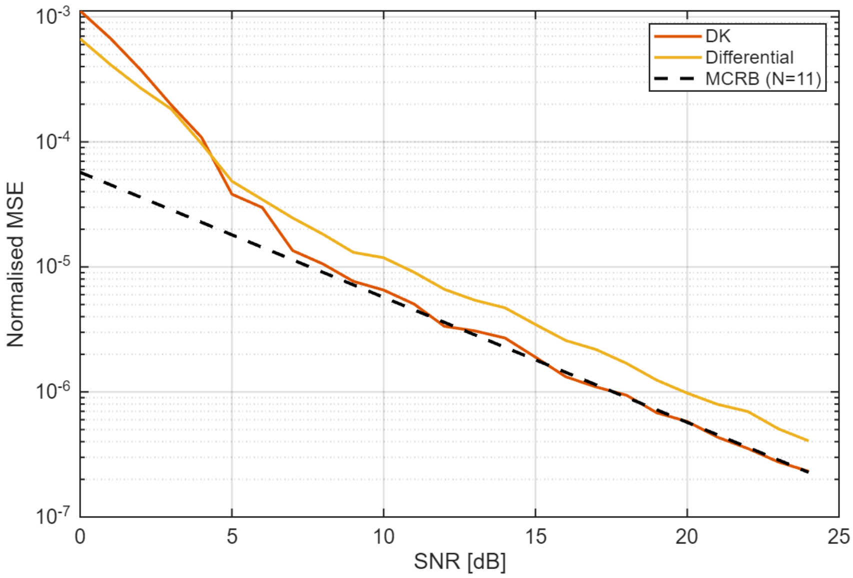


Figure 3.3: Normalised MSE of frequency estimation with the DK estimator for $L = 11$ training symbols.

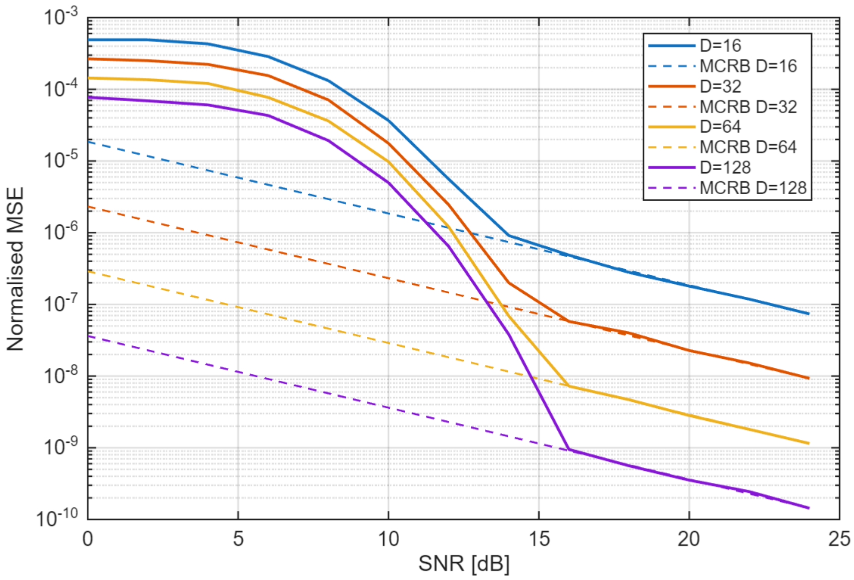


Figure 3.4: Normalised MSE of the DK estimator with blind operation for different observation lengths.

3.2.4 Computational Complexity

Here, the computational complexity of both frequency recovery algorithms is compared in terms of real multiplications and real additions. CORDIC is used to find phases and is assumed to cost the same as a real multiplication. Real division is also assumed to cost the same as multiplication since it is only performed once (this would be a suitable approximation if look-up tables are used to find reciprocals at some cost to accuracy). All analysis is for data-aided operation, avoiding the fourth power operation in eq. (3.1), and applies to a single polarization. If averaging is done over both polarizations, then all operations are doubled.

R&B The R&B algorithm requires an FFT followed by N_{FFT} comparisons of the magnitudes in the coarse search. Many of the FFT entries are 0 due to padding, which reduces the total number of operations needed. The interpolation requires the real part of a complex division

$$\Re\left[\frac{a+bj}{c+dj}\right] = \frac{ac+bd}{c^2+d^2}. \tag{3.9}$$

The total complexity is summarised in table 3.1, with the complexity for blind operation marked with (*).

Operation	RM	RA
Forming $z[i], \ldots, z[i+L]$	$4L$, $12L^*$	$3L$, $9L^*$
FFT	$2L\log_2(\frac{N_{FFT}}{L}) + 2N_{FFT}\log_2(L)$	$3L\log_2(\frac{N_{FFT}}{L}) + 3N_{FFT}\log_2(L)$
Search	$2N_{FFT}$	$2N_{FFT}$
Interpolation	5	4

Table 3.1: Computational complexity of estimating CFO with the R&B algorithm.

DK The DK estimator has lower complexity, only requiring CORDIC, real additions, and real multiplications. The fixed multiplications by the weights of the Kay estimator can be optimised into a low-energy shift-add circuit, but this is ignored for simplicity. The first-pass CFO correction involves forming the phase corrections of each training symbol as $\phi[i] = \hat{f}_d i$, then applying them with CORDIC.

Operation	RM	RA
Forming $\arg(z[i]), \ldots, \arg(z[i+L])$	$2L$	L
Differential Phase	1	$L-1$
First Pass Correction	$2L$	0
Forming $\arg(z[i]), \ldots, \arg(z[i+L])$	$2L$	L
Kay	L	$L-1$

Table 3.2: Computational complexity of estimating CFO with the DK estimator.

3.2.5 Optimising Bit Width

These frequency recovery algorithms compute an estimate of the normalised CFO f_d/R_s. For n bits of precision, the smallest normalised CFO that can be estimated is 2^{-n} for an estimation range $0 \leq |f_d/R_s| \leq 1$. In reality, a much smaller range of CFOs needs to be estimated. If the maximum permitted value of $|f_d/R_s| = 2^{-d}$, d bits of precision that would otherwise be redundant can be utilised by scaling each estimate by 2^d during calculation, then inserting d bits in the final answer to recover the true normalised CFO. This improves the resolution to $2^{-(n+d)}$ without any increase in energy consumption.

3.2.6 Maximum CFO

The theoretical limit on the maximum CFO that can be corrected is given by Nyquist as $0.5R_s$. From eq. (3.1), we see that blind operation reduces this limit by a factor of 4. Both the R&B and DK estimators accurately estimate the CFO up to the Nyquist limit, as seen in fig. 3.5. Blind operation aliases as expected at $f/R_s = 0.125$, making it viable for CFOs up to 3.8 GHz with a symbol rate of 30.5 GHz.

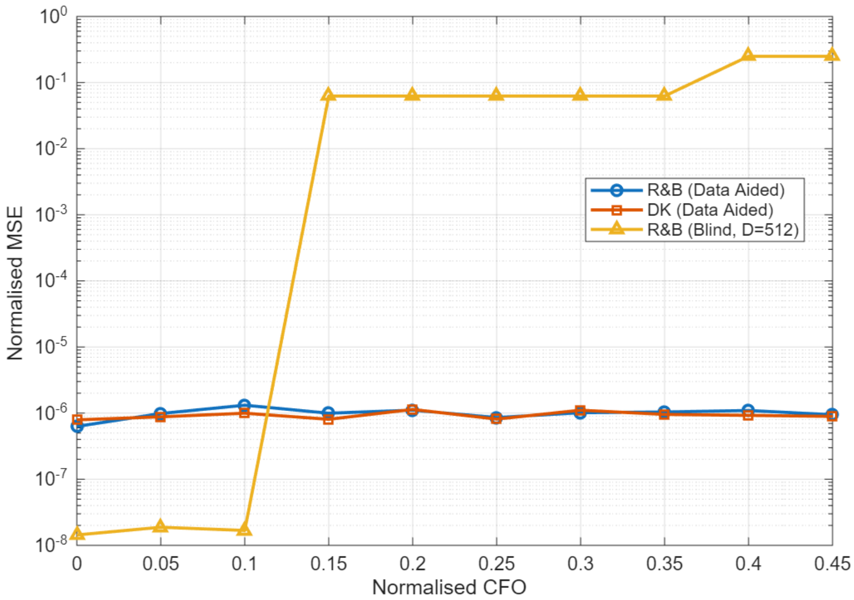


Figure 3.5: Normalised MSE in CFO estimate (MSE^2/R_s^2) vs Normalised CFO (f/R_s)

3.3 Phase Recovery

3.3.1 Feed-Forward vs PLL Algorithms

One of the first techniques investigated for phase recovery was a digital phase-locked loop (PLL) to incrementally correct the phase of each symbol with feedback. This dramatically simplifies the phase estimation and correction due to the use of small-angle approximations. The CPON specification sets the symbol rate at 30.5 GBd. With an oversampling rate of 8/7, the total samples per second would be 35 GSa/s. This is beyond the achievable clock speed of CMOS, so symbols must be processed in blocks of e.g., 32 symbols with a clock speed of 1.1 GHz. This approach hinders the effectiveness of a PLL, as updates can only be made every 32 symbols. For this reason, feed-forward algorithms that can operate on blocks of data were used instead.

3.3.2 Viterbi & Viterbi

The Viterbi & Viterbi algorithm is well suited to QPSK and removes data phase with the fourth power operation of eq. (3.1). It then applies a maximum-likelihood filter $w_{ML}[i]$ to estimate the phase error due to Wiener phase noise from the combined laser linewidth and any residual CFO as

$$\hat{\theta}_{ML}[i] = \frac{1}{4} \arg\left(\sum_{k=1}^{N} w^*_{ML}[k] z[k]\right) - \frac{\pi}{4}. \tag{3.10}$$

A phase unwrapper is then applied before the phase error is corrected. This is readily adapted to block-based processing. If blocks of 32 symbols are used again, the ML filter can be applied to all symbols in the block and the estimated phase error applied to the entire block. This is only valid if the phase difference between the start and end of the block is small, meaning the laser linewidth cannot be too large and the preceding frequency recovery algorithm must correct most of the CFO. By increasing the time between phase estimates, block processing also increases the chance of cycle-slips in the phase unwrapper. The CPON specification provides 1 pilot symbol in every 32-symbol block that can be used to correct this [2]. The correction estimates the number of $\pi/2$ phase jumps n that occur between the unwrapped phase of a data symbol $\hat{\theta}_{PU}$ and its corresponding pilot, then adjusts the unwrapped estimate as $\hat{\theta}'_{PU} = \hat{\theta}_{PU} - n\frac{\pi}{2}$.

3.3.3 Pilots-Only Correction

While Viterbi & Viterbi is commonly used for QPSK, a simpler approach is to compute the phase error from the pilot symbol using eq. (3.2) and apply this to the whole block. This also relies on a small phase change across the block and is more vulnerable to additive noise. However, QPSK allows for $\pm 45°$ error in the phase estimate before an incorrect decision is made, which makes it robust to the drop in accuracy associated with this approach.

3.3.4 Computational Complexity

Viterbi & Viterbi For a block-based approach, the length of the filter will be the block length. Phases are again found with CORDIC. The cost of the phase estimate for each block is shown in table 3.3.

Operation	RM	RA
Forming $z[i], \ldots, z[i+N]$	$12N$	$9N$
Computing $\hat{\theta}_{ML}$	$4N+1$	$3N+2(N-1)+1$

Table 3.3: Computational complexity of block-based Viterbi & Viterbi.

Pilots-Only Only phases are needed, so the entire estimation can be performed with CORDIC. The pilot phase is also known beforehand, so does not contribute to the computational cost. The total cost of the phase estimate is shown in table 3.4.

Operation	RM	RA
Forming $\arg(z[i]) = \hat{\theta}$	1	1

Table 3.4: Computational complexity of Pilots-Only correction.

3.4 Methodology

3.4.1 Phase Recovery Comparison

The two phase recovery algorithms are compared in isolation by simulating an AWGN channel with additive Wiener phase noise, without chromatic dispersion, PMD, or fibre nonlinearity, so that any performance difference is attributable solely to the phase estimation stage. BER is estimated by Monte Carlo simulation over independent realisations for each combination of laser linewidth and SNR, with the phase ambiguity inherent to QPSK resolved by exhaustive search over the four $\pi/2$ rotations before symbol decisions are made.

3.4.2 System Energy Comparison

The energy framework of chapter 2 is applied to compare all combinations of frequency and phase recovery algorithm. All simulations target the CPON specification: DP-QPSK at $R_s = 30.5$ GBd with a pre-FEC BER limit of 2×10^{-2} and a laser wall-plug efficiency $\eta = 0.1$. Two link scenarios are considered, with parameters given in table 3.5, for splitting ratios $K \in \{1, 10, 20\}$.

Table 3.5: Link parameters for the two noise regimes.

Parameter	Symbol	Value
Shot-noise limited (10 km passive-split PON)		
Fibre attenuation	α	0.2 dB/km
Link length	L	10 km
Amplified (3× 80 km EDFA spans)		
Number of spans	n	3
Span length	L_{span}	80 km
Fibre attenuation	α	0.2 dB/km
Group velocity dispersion	$\lvert\beta_2\rvert$	$20 \times 10^{-24}\,\text{s}^2/\text{km}$
Nonlinear coefficient	γ	$1.3\,\text{W}^{-1}\,\text{km}^{-1}$
EDFA noise figure	NF	5 dB

For each algorithm combination, the fixed-point word lengths of the frequency recovery (n_{FR}) and phase recovery (n_{PR}) stages are varied independently over a grid. At each grid point, Monte Carlo simulation over multiple independent realisations estimates the minimum SNR at which the pre-FEC BER falls below the CPON threshold; this FEC SNR threshold determines the required transmitter power via eqs. (2.5) and (2.10). The receiver DSP energy at that precision is computed from the operation counts of the preceding sections via eq. (2.4), and the transmitter energy follows from eq. (2.7). The precision combination that minimises the mean system energy of eq. (2.11) is then selected for each algorithm pair and splitting ratio.

The reported energies should be interpreted as a *comparative* metric. The 45 nm CMOS reference values of table 2.1 provide a consistent basis for ranking implementations, but do not constitute absolute power predictions for a real device. Additionally, the energy cost of applying the estimated corrections to each received symbol — CORDIC rotations shared by all implementations — is excluded, as this contribution is implementation-independent and does not affect relative comparisons.

3.5 Results & Discussion

3.5.1 Phase Recovery Performance

To replace Viterbi & Viterbi with the simplified pilots-only phase correction, it must exhibit comparable performance over the range of linewidths that could be expected during operation of the network. The BER for each phase recovery algorithm as SNR varies is shown in fig. 3.6. Both algorithms have almost identical performance and are reliable up to 10 MHz, well beyond the maximum specified in the CPON spec.

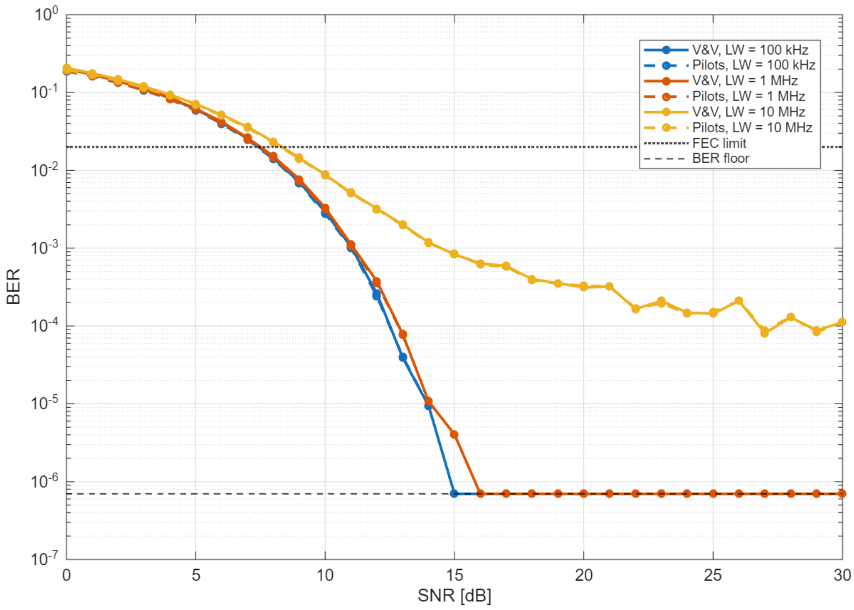


Figure 3.6: Performance comparison of V&V and Pilots-Only phase correction

3.5.2 Blind vs Data-Aided Operation

The decrease in threshold SNR for longer observation lengths suggests that the performance of the R&B algorithm will improve with blind observation length D. This is not the case for the DK estimator, where increasing blind observation length has no effect on threshold SNR. The effect of observation length on FEC SNR is shown for both algorithms in fig. 3.7, where blind R&B shows a clear increase in performance over the blind DK estimator for longer observations and improves on data-aided R&B after ≈ 50 symbols. The DK estimator in blind operation is thus not viable for frequency recovery, but blind R&B can provide the best performance at a higher computational cost.

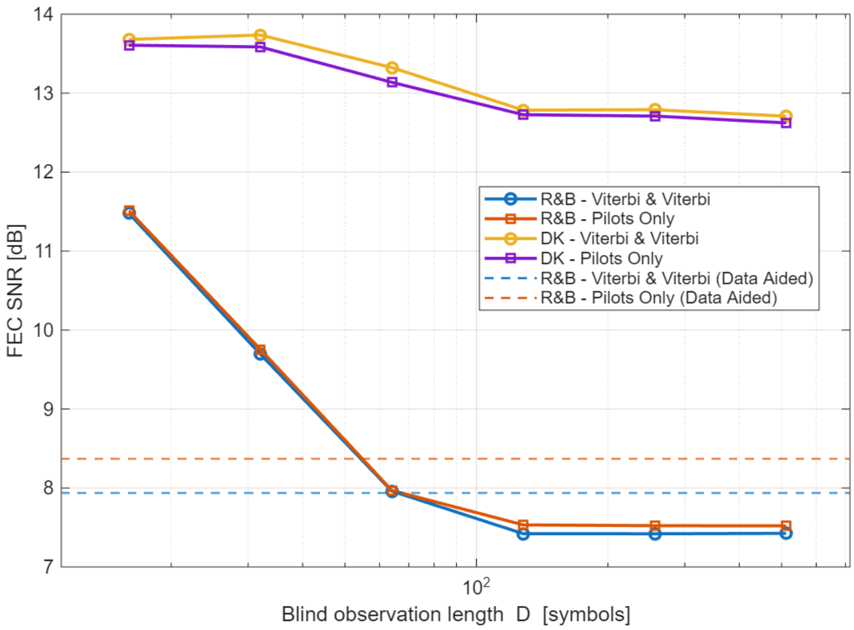
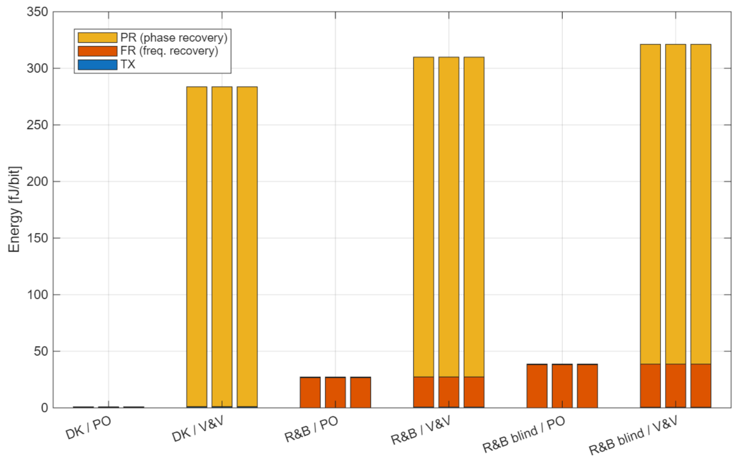


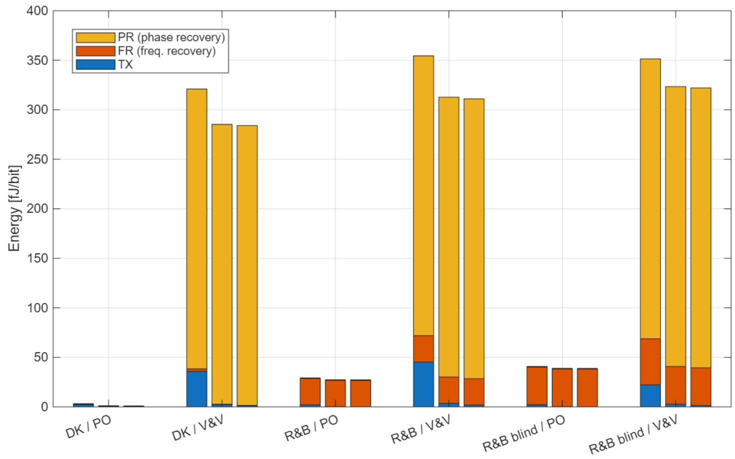
Figure 3.7: FEC SNR vs blind observation length for the R&B and DK estimators. CFO = 2 GHz, laser linewidth = 1 MHz.

3.5.3 Combined Performance

The lowest energy consumption required to achieve the FEC limit for each combination of frequency and phase recovery algorithm is compared in fig. 3.8. The choice of phase recovery algorithm dominates the change in total energy, with Viterbi & Viterbi consuming approximately 300 fJ/bit more than Pilots-Only correction. Amortisation over more symbols means that frequency recovery is less significant, with blind R&B only contributing approximately 40 fJ/bit more than DK. Transmitter energy is negligible in the shot-noise regime, and is only significant for $K = 1$ under amplifier and nonlinear noise. The exact energy consumption of each implementation and the corresponding FEC SNR are shown in table 3.6. The high contribution from receiver energy is reflected in the low number of bits used for fixed-point arithmetic, where the increase in computational cost from higher precision outweighs any gains in transmitter efficiency from a reduced FEC SNR. The DK estimator with Pilots-Only phase correction offers the lowest combined energy at a comparatively low FEC SNR. However, if the CFO remains constant over a much longer period, then the estimate can be applied over more symbols and the per-symbol energy can be neglected. This favours the high performance offered by blind R&B, making it more efficient than DK as shown in table 3.7.



(a) Shot-noise limited (10 km passive-split PON).



(b) Amplified link (3×80 km EDFA spans).

Figure 3.8: System energy split by component for each FR/PR algorithm combination and splitting ratio K (left: $K = 1$, middle: $K = 10$, right: $K = 20$).

Table 3.6: Minimum combined per-bit energy for the transmitter and carrier recovery, in fJ/bit, for the two link regimes.

(a) 10 km passive-split PON (shot-noise limited).

FR	PR	K	$E_{\text{sys}} \pm \sigma$ (fJ/bit)	FEC SNR (dB)	$(n_{\text{FR}}, n_{\text{PR}})$
DK	Pilots	1	$\mathbf{0.922 \pm 0.0044}$	9.76 ± 0.89	$(2, 2)$
		10	$\mathbf{0.922 \pm 0.0044}$	9.76 ± 0.89	$(2, 2)$
		20	$\mathbf{0.922 \pm 0.0044}$	9.76 ± 0.89	$(2, 2)$
	V&V	1	284 ± 0.64	23.2 ± 5.4	$(2, 2)$
		10	284 ± 0.64	23.2 ± 5.4	$(2, 2)$
		20	284 ± 0.64	23.2 ± 5.4	$(2, 2)$
R&B	Pilots	1	27.2 ± 0.0026	9.30 ± 0.61	$(2, 2)$
		10	27.2 ± 0.0026	9.30 ± 0.61	$(2, 2)$
		20	27.2 ± 0.0026	9.30 ± 0.61	$(2, 2)$
	V&V	1	310 ± 0.55	23.6 ± 4.2	$(2, 2)$
		10	310 ± 0.55	23.6 ± 4.2	$(2, 2)$
		20	310 ± 0.55	23.6 ± 4.2	$(2, 2)$
R&B blind (D = 32)	Pilots	1	38.7 ± 0.0028	9.40 ± 0.63	$(2, 2)$
		10	38.7 ± 0.0028	9.40 ± 0.63	$(2, 2)$
		20	38.7 ± 0.0028	9.40 ± 0.63	$(2, 2)$
	V&V	1	321 ± 0.48	22.1 ± 4.4	$(2, 2)$
		10	321 ± 0.48	22.1 ± 4.4	$(2, 2)$
		20	321 ± 0.48	22.1 ± 4.4	$(2, 2)$

(b) 3×80 km amplified link (ASE + NLI limited).

FR	PR	K	$E_{\text{sys}} \pm \sigma$ (fJ/bit)	FEC SNR (dB)	$(n_{\text{FR}}, n_{\text{PR}})$
DK	Pilots	1	$\mathbf{3.19 \pm 0.51}$	9.76 ± 0.89	$(2, 2)$
		10	$\mathbf{1.13 \pm 0.051}$	9.76 ± 0.89	$(2, 2)$
		20	$\mathbf{1.02 \pm 0.026}$	9.76 ± 0.89	$(2, 2)$
	V&V	1	321 ± 29	20.2 ± 3.9	$(6, 2)$
		10	285 ± 1.8	18.6 ± 3.3	$(2, 2)$
		20	284 ± 0.94	18.6 ± 3.3	$(2, 2)$
R&B	Pilots	1	29.2 ± 0.31	9.30 ± 0.61	$(2, 2)$
		10	27.4 ± 0.031	9.30 ± 0.61	$(2, 2)$
		20	27.3 ± 0.015	9.30 ± 0.61	$(2, 2)$
	V&V	1	354 ± 28	21.7 ± 3.3	$(2, 2)$
		10	313 ± 2.1	20.6 ± 3.0	$(2, 2)$
		20	311 ± 1.1	20.6 ± 3.0	$(2, 2)$
R&B blind (D = 32)	Pilots	1	40.8 ± 0.33	9.40 ± 0.63	$(2, 2)$
		10	38.9 ± 0.033	9.40 ± 0.63	$(2, 2)$
		20	38.8 ± 0.016	9.40 ± 0.63	$(2, 2)$
	V&V	1	351 ± 19	18.4 ± 3.5	$(2, 2)$
		10	323 ± 2.0	19.3 ± 3.0	$(2, 2)$
		20	322 ± 1.0	19.3 ± 3.0	$(2, 2)$

Table 3.7: Minimum combined per-bit energy when the frequency recovery energy is amortised away.

(a) 10 km passive-split PON (shot-noise limited).

FR	PR	K	$E_{\text{sys}} \pm \sigma$ (fJ/bit)	FEC SNR (dB)	$(n_{\text{FR}}, n_{\text{PR}})$
DK	Pilots	1	0.592 ± 0.0022	8.84 ± 0.59	$(14, 2)$
		10	0.592 ± 0.0022	8.84 ± 0.59	$(14, 2)$
		20	0.592 ± 0.0022	8.84 ± 0.59	$(14, 2)$
R&B blind (D = 512)	Pilots	1	$\mathbf{0.590 \pm 0.00049}$	8.39 ± 0.15	$(12, 2)$
		10	$\mathbf{0.590 \pm 0.00049}$	8.39 ± 0.15	$(12, 2)$
		20	$\mathbf{0.590 \pm 0.00049}$	8.39 ± 0.15	$(12, 2)$

(b) 3×80 km amplified link (ASE + NLI limited).

FR	PR	K	$E_{\text{sys}} \pm \sigma$ (fJ/bit)	FEC SNR (dB)	$(n_{\text{FR}}, n_{\text{PR}})$
DK	Pilots	1	2.40 ± 0.26	8.84 ± 0.59	$(14, 2)$
		10	0.759 ± 0.026	8.84 ± 0.59	$(14, 2)$
		20	0.667 ± 0.013	8.84 ± 0.59	$(14, 2)$
R&B blind (D = 512)	Pilots	1	$\mathbf{2.21 \pm 0.057}$	8.39 ± 0.15	$(12, 2)$
		10	$\mathbf{0.739 \pm 0.0057}$	8.39 ± 0.15	$(12, 2)$
		20	$\mathbf{0.658 \pm 0.0029}$	8.39 ± 0.15	$(12, 2)$

3.6 Conclusions

The robustness to both additive and phase noise of QPSK allows the phase recovery block of a coherent receiver to be dramatically simplified by just using pilot symbols with no noticeable drop in performance, resulting in considerable energy savings. Frequency recovery can also be implemented with simple algorithms to save energy; however, the efficacy of doing so depends on how often the CFO must be estimated. The difference in energy consumption between different carrier recovery implementations is dominated by the receiver, and the optimal implementation does not seem to depend on the noise regime of the network.

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